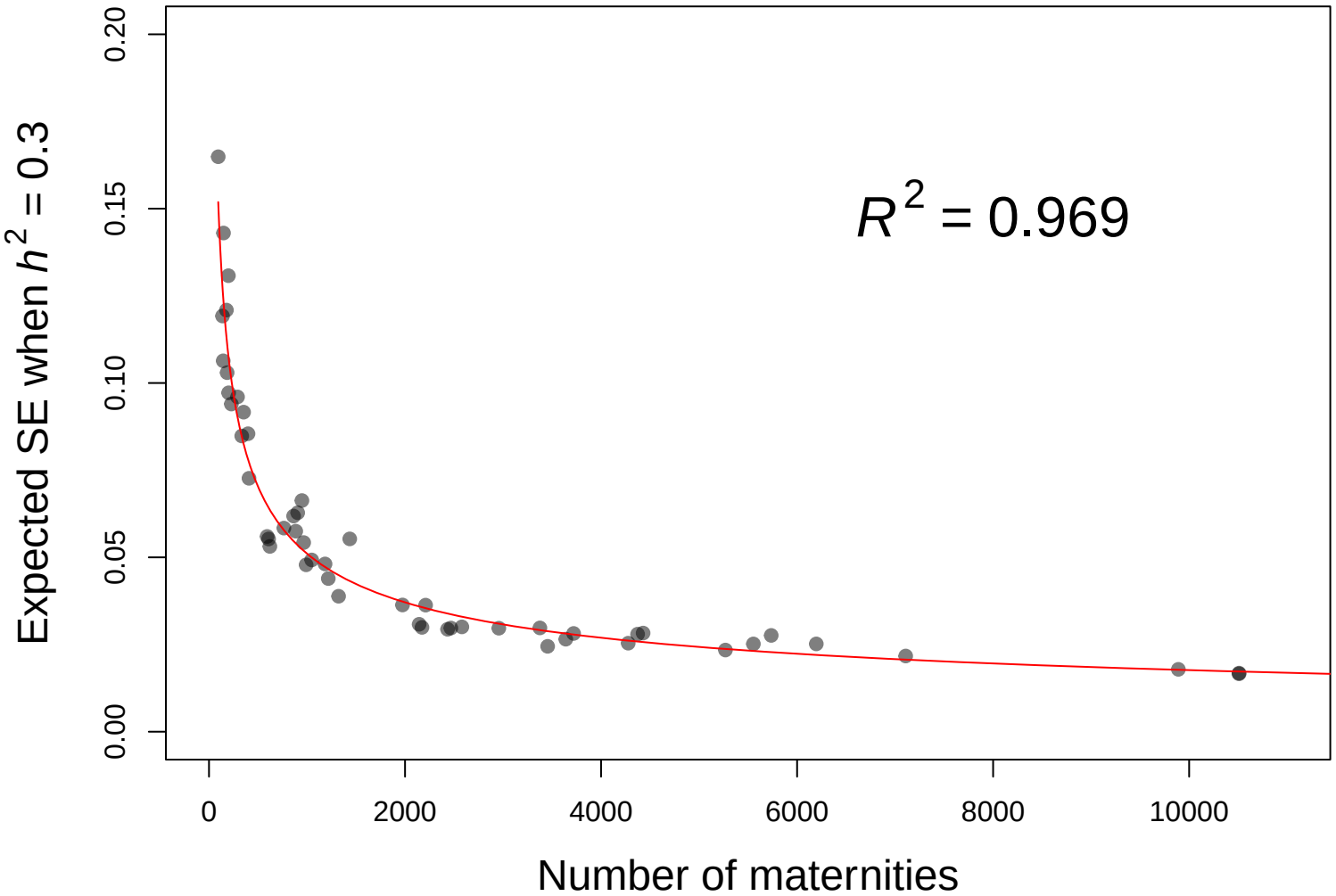
a)

Metric	R^2
records	0.917
maternities	0.969
paternities	0.923
full sibs	0.642
maternal sibs	0.732
maternal half sibs	0.51
paternal sibs	0.667
paternal half sibs	0.465
maternal grandmothers	0.445
maternal grandfathers	0.464
paternal grandmothers	0.474
paternal grandfathers	0.436
maximum pedigree depth	0.097
founders	0.472
mean maternal sibship size	0.277
mean paternal sibship size	0.204
non−zero F	0.082
F > 0.125	0.016
mean pairwise relatedness	0.147
pairwise relatedness>=0.125	0.165
pairwise relatedness>=0.25	0.271
pairwise relatedness>=0.5	0.409
var pairwise relatedness	0.277

b)